

Vitthal Mirji

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SUMMARY

- **Staff/Principal Data Architect:** 12+ years building contract-driven, developer-friendly data platforms for Walmart US, Walmart International and Deloitte Fortune 500 clients. I partner with product to define what to build (experience ecosystems, monetization surfaces, observability) and lead how to build using functional Scala, Spark/Databricks, Kafka/Flink, and Google Cloud. Targeting Staff Data Architect / Lead Data Platform roles across Germany, France, Netherlands, Finland or the UK. Community-first: creator of flowforge & toon4s, core contributor to llm4s, cask, scalasql, international speaker (ScalaIO/Scala Days), and bootcamp instructor.
- **Also open to Scala Engineering roles:** Keen on functional programming, open-source contribution, clean code and software craftsmanship, using modern effect systems (Cats Effect, ZIO) and production-grade FP practices.

KEY ACCOMPLISHMENTS

- **Framework adoption at scale:** Shipped six reusable frameworks (data-contracts, Scala/Spark/data-engineering archetypes, monitoring, QA, Data & ML Ops) that cut delivery time from six weeks to two and reduced incidents by 82% across 40+ teams.
- **Monetization impact:** Monetized closed-loop merchant insights processing 100M+ daily records on Spark/Databricks, unlocking 75% business growth with sub-2h freshness.
- **Cost and SLA gains:** Modernized Deloitte estates (Aflac, Marriott, Eli Lilly), trimming infra spend by 62% (US\$2.4M→US\$900K) and boosting SLAs to 99.5%.
- **Data quality at scale:** Built DPAT automation + data-quality engines validating 5TB+/day at 99.9% accuracy, shrinking QA sign-off from three days to four hours.
- **Open source:** flowforge & toon4s creator, llm4s core contributor/GSoC mentor, full-stack Scala contributions (cask, scalasql).
- **Talks & training:** ScalaIO/Scala Days/Google Developer Groups; Java/Kotlin user groups speaker; bootcamp Scala/data-engineering trainer.

CORE SKILLS

- **Architecture:** Data contracts, Data Engineering design platforms, design patterns, streaming observability, data governance, monetization products, code-generation platforms, data warehousing, data modeling, data quality, low-level design, OOP, functional programming, SOLID principles, system design, TDD.
- **Languages & Engines:** Scala (Typelevel effects, FP), Python, Rust, SQL, Apache Spark, Databricks, Delta Lake, Flink, Kafka, Beam.
- **Cloud:** GCP (BigQuery, Dataproc, GCS, Pub/Sub), AWS (S3, Glue), Docker.
- **DevOps & Orchestration:** Airflow, Git, Jenkins, Maven, SBT, Mill, Docker, CI/CD, GitHub Actions.
- **Leadership:** Staff+ design reviews, roadmap definition, cross-team adoption playbooks, senior IC mentoring, incident/SLO programs, Agile/Scrum.

EXPERIENCE

- **Walmart Global Tech** Bangalore, India
Staff Data Engineer (IC5 / SDE5 Architect) Nov 2023 – Present
 - **Product to architecture:** Shape merchant analytics and supplier collaboration strategy; translate product bets into architecture and adoption roadmaps.
 - **Framework suite:** Contracts, archetypes, monitoring, QA, Data & ML Ops frameworks powering 40+ teams / 150+ engineers; 70% less boilerplate.
 - **DPAT & observability:** Automation plus real-time observability cut QA sign-off from 3 days to 4 hours and MTTR by 60%.
 - **Mentorship:** Mentor eight senior engineers on functional Scala, ADRs, and staff-level readiness.

- Walmart Global Tech** Bangalore, India
Senior Data Engineer / Tech Lead May 2021 – Oct 2023
 - **Closed-loop insights:** Led 13-engineer squad delivering sales/inventory/supply-chain insights over 100M+ daily records with 12h freshness.
 - **Reusable accelerators:** Scala/Python APIs and CI/CD templates adopted by 12 teams for rapid domain onboarding.
 - **Supplier latency:** Improved supplier collaboration latency from 48h to 2h; forecasting accuracy +35% via streaming connectors.
 - **Cost & quality:** Optimized Spark/Dataprocs cost by 40% while enforcing data-quality SLAs.
- Deloitte Consulting (U.S. offices)** Mumbai, India
Specialist Senior - Data Engineering Apr 2017 – May 2021
 - **Enterprise EDW:** Architected 350+ table EDW for Fortune 500 clients, increasing engagement revenue by 43%.
 - **Cloud migration:** Moved 100+ TiB Hadoop workloads to AWS S3/Glue with zero data loss and automated reconciliations.
 - **ETL libraries:** Scala/Python ETL libraries ensuring 99.9% accuracy; launched governance and SLA dashboards.
 - **Leadership:** Led pods of 6–8 engineers covering hiring, mentoring, and performance calibrations.
- Bista Solutions Inc.** Mumbai, India
Senior Data Engineer & Analyst Feb 2016 – Apr 2017
 - **Predictive demand:** Designed predictive demand platform for U.S. automotive clients (+5% sales, -10% inventory, +60% forecast accuracy).
 - **Streaming pipelines:** Built Spark Streaming + Kafka pipelines for real-time supply-chain telemetry.
- Nihilent Technologies** Pune, India
Data Engineer Nov 2015 – Feb 2016
 - **Ingestion accelerators:** Created bash-based ingestion accelerators importing 1,000+ tables into Hadoop across 50+ programs.

OPEN SOURCE

Author and core contributor to Scala data engineering libraries and AI frameworks spanning compile-time safety, functional pipelines, and LLM tooling.

- **flowforge:** Type-safe Scala data-engineering toolkit with compile-time contract validation; prevents schema drift pre-deploy. [GitHub](#)
- **llm4s:** Agentic LLM programming framework for Scala with multi-provider support; core contributor and GSoC mentor. [GitHub](#)
- **datapipelines-essentials-python:** Simplified Spark ETL with SparkSession extensions, DataFrame validation, SQL functions. [GitHub](#)
- **toon4s:** Token-Oriented Object Notation for JVM; reduces LLM token costs by 30–60% while keeping JSON compatibility. [GitHub](#)
- **compile-time-data-contracts:** Scala 3 macro-based schema validation; pipelines fail to compile on schema drift. [GitHub](#)
- **spark-spring-boot:** Apache Spark with Spring Boot for rapid prototyping and enterprise integration. [GitHub](#)
- **scalasql:** Contributor to Li Haoyi's type-safe SQL library; added COUNT and COUNT(DISTINCT) aggregation. [GitHub](#)
- **cask:** Contributor to Li Haoyi's Scala web framework; added DB transaction support and app configuration. [GitHub](#)
- **Rock the JVM Articles:** Author of Scala articles. [GitHub](#)
- **abortable-bytes:** Kyo effect-system POC for cancel-safe cloud uploads. [GitHub](#)
- **spark-xml:** Contributor to Databricks' Spark XML library. [GitHub](#)
- **Data Pipelines Simplified:** Contributor to metadata framework.

TALKS & SPEAKING

- **Scala.io Paris 2025:** Compile-time data contracts & fiber-safe data pipelines: Scala's effect system in action. ([Scala.io](#))
- **OAISYS AI Practitioners 2025, Pune:** LLM4S: Build reliable AI systems in the JVM ecosystem.
- **Scala India 2025:** When data pipelines fail early: type-safe data engineering in Scala.
- **Scala Days:** Speaker on Scala functional data engineering. ([Scala Days](#))
- **GDGs & JUGs:** Regular talks at Google Developer Groups/Java-Kotlin user groups on Scala, Spark, data architecture.
- **Bootcamps:** Internal training on data engineering & functional programming for Fortune 500 teams.
- **Technical articles:** Publish on [vitthalmirji.com](#) and RockTheJVM.

EDUCATION

- **Visvesvaraya Technological University** Belgaum, India
B.E. Computer Science (First Class) *Jun 2015*
- **Bangalore University** Bangalore, India
Diploma Computer Science (First Class Distinction) *2010*

CERTIFICATIONS

- **Google Professional Data Engineer:** Feb 2021
- **Cloudera Certified Spark & Hadoop Developer:** May 2020
- **Sun Certified Java Developer:** Java 5.x (2009), 6.x (2010)

RECOGNITION

- **Walmart Bravo Award:** 2024 & 2021
- **Walmart Wall of Fame:** 2021–2024
- **Deloitte Outstanding Performance:** 2019
- **Best Academic Project:** 1st/15, 2015, Gogte Institute of Technology